Assignment 2

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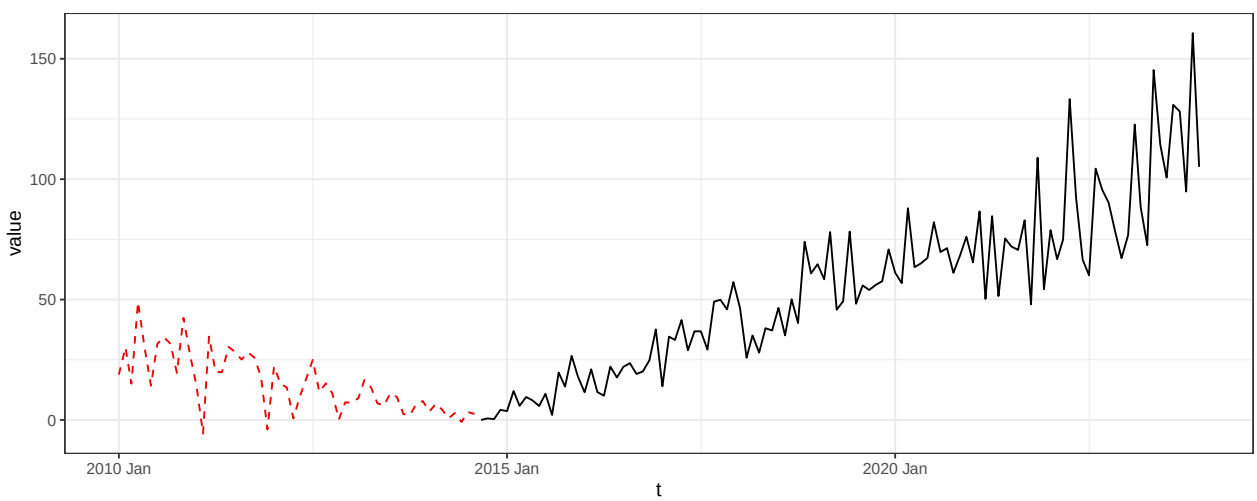
```
theme_set(theme_bw())
```

I. Assessing Trend

- Assess the need for a trend component and include it in the model, as you see fit. I*f needed, reduce the data to model only those observations for which the trend is consistent. Note that, if you decide to reduce the data, to be consistent from here on you must work with the reduced data.*

Series 1

```
data1.tibble <- readr::read_csv("assignment_1-2024-data1.csv", show_col_types = F)
colnames(data1.tibble) <- c('t', 'value')
data1 <- data1.tibble %>%
mutate(t = yearmonth(t)) %>%
as_tsibble(index = t)
ggplot() +
geom_line(aes(x = t, y = value), data = data1 %>% slice(1:56), lty = 2, col = 'red2') +
geom_line(aes(x = t, y = value), data = data1 %>% slice(57:n()))
```

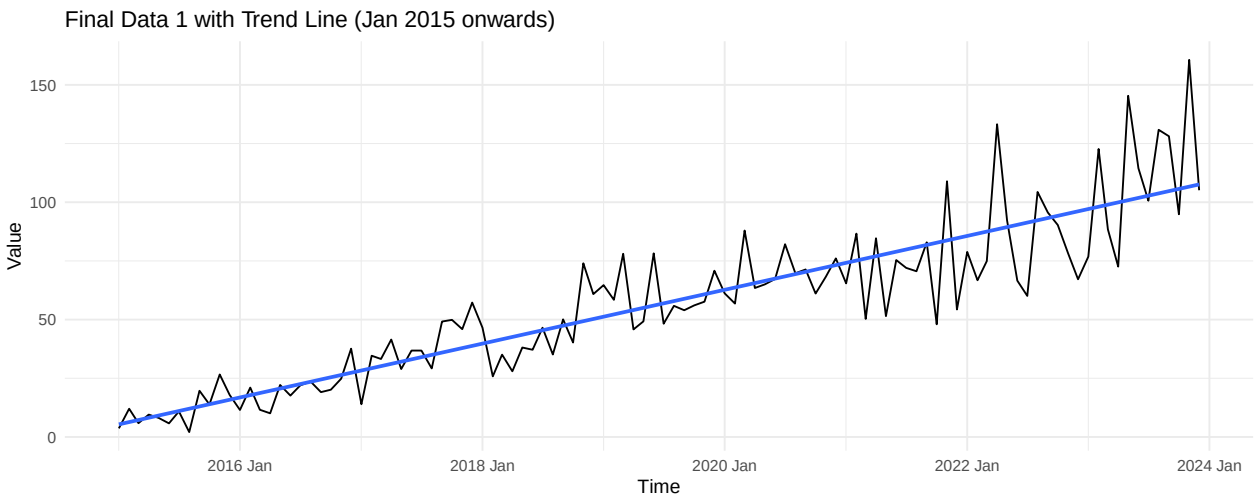


For `data1`, an upwards trend is visible from around January 2015 onward. For further analysis, exploration and modelling, the data set is modified and will no longer consider data prior to year 2015. Below, the condensed data1_final graph is displayed, alongside a trend line, indicating this upward trend.

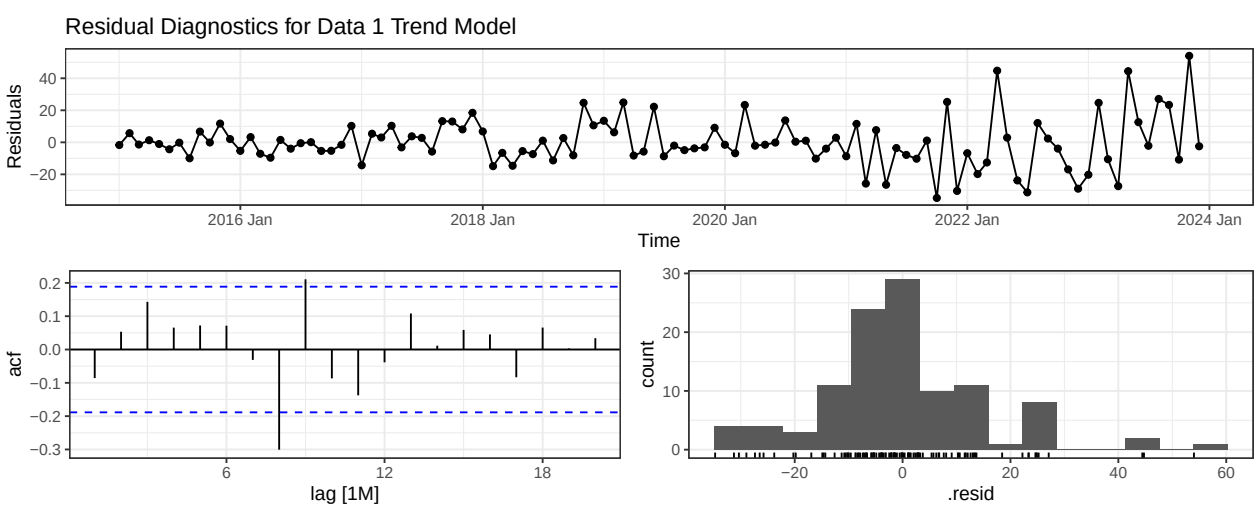
```
data1_final <- data1 %>% filter(t >= yearmonth("2015 Jan"))
```

```
rm(list = setdiff(ls(), c("data1_final", "data2")))
```

```r
data1_final %>% ggplot(aes(x = t, y = value)) +
  geom_line() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Final Data 1 with Trend Line (Jan 2015 onwards)", x = "Time", y = "Value", linetype = "
  theme_minimal()
```




```r
data1_trend_model <- data1_final %>% model(TSLM(value ~ trend()))

gg_tsresiduals(data1_trend_model) +
  labs(title = "Residual Diagnostics for Data 1 Trend Model",
       x = "Time", y = "Residuals")
```



Starting with the trend model residuals, the model displays that the distance between fitted values seem to average out to 0 and do not follow a clear pattern, with variance increasing from 2022 to 2024. Based on this graph alone, one can conclude that the model fits the data quite well. Notably, the increasing variance towards later years potentially indicates heteroskedasticity, which the model fails to capture in that period.

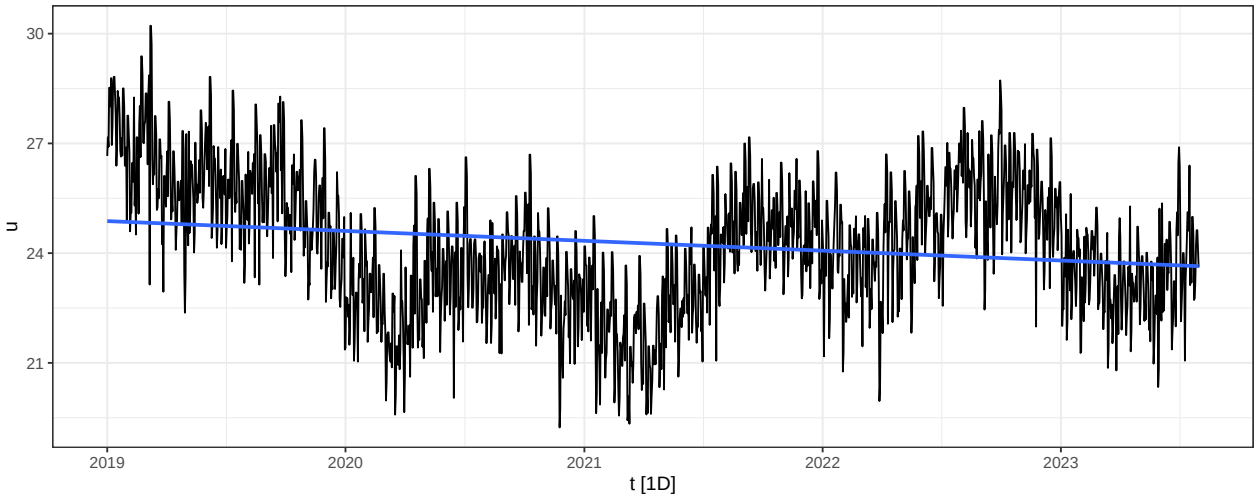
The ACF of the residuals displays two significant values at lag 8 and 9, but no further pattern. This suggests that the model has room for improvement as there may still be some unexplained structure in the data. However, a broader pattern cannot be identified indicating that the model captures most of the systematic components of the data.

The distribution of residuals follows an almost normal distribution (bell shaped), a characteristic of a good

model.

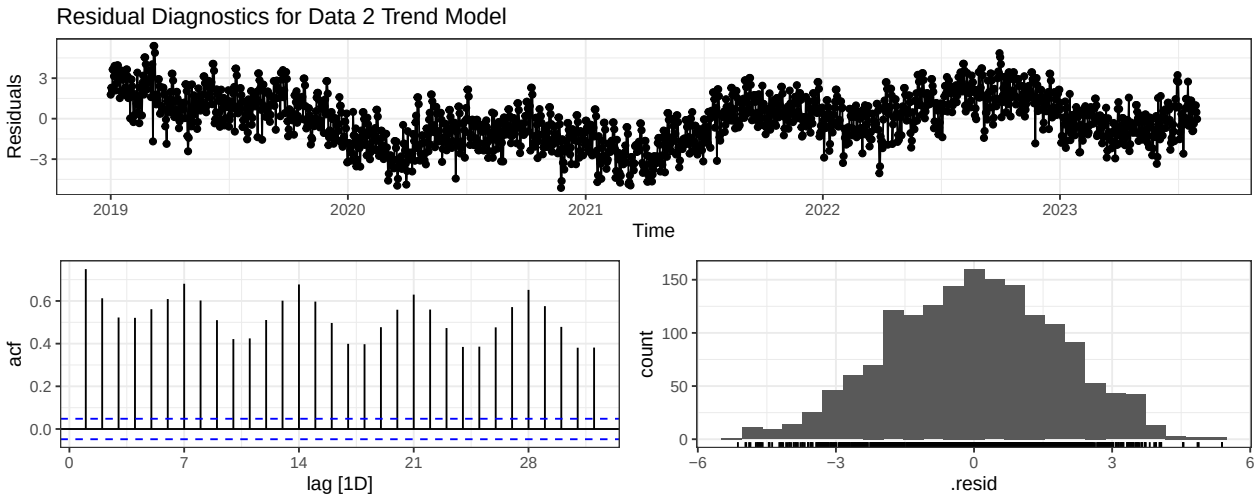
Series 2

```
data2 <- readr::read_csv("assignment_1-2024-data2.csv", show_col_types = F) %>%
  as_tsibble(index = t)

data2 %>%
  autoplot(u) +
  geom_smooth(method = lm, formula = y ~ x, se = FALSE)
```



A slight downward trend can be spotted for `data2`. Subsequent residual analysis is necessary to explore the need and impact of incorporating trend in a model.

```
data2_trend_model <- data2 %>% model(TSLM(u ~ trend()))
data2_trend_model.augmented <- data2_trend_model%>%
  augment()

gg_tsresiduals(data2_trend_model)+
    labs(title = "Residual Diagnostics for Data 2 Trend Model",
       x = "Time", y = "Residuals")
```



The residuals for `data2`are dispersed around zero but seem to follow a systematic pattern over time.

The ACF of the residuals displays significant values at each of the 30 lags, with a clear seasonality pattern. With spikes every 7 lags, this could indicate weekly seasonality. This means that the trend model may be missing a seasonal component and could benefit from incorporating one.

The histogram of the residuals is bell shaped and normally distributed around 0, which is a positive aspect of the model. However, the other two residual plots indicate that the trend model would require a seasonal component to improve.

```
data2_trend_model%>%
  report()
```

```
## Series: u
## Model: TSLM
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.12892 -1.28623  0.04609  1.27599  5.38747
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.488e+01  8.954e-02 277.816  < 2e-16 ***
## trend()     -7.375e-04  9.266e-05  -7.959 3.18e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 1671 degrees of freedom
## Multiple R-squared: 0.03652, Adjusted R-squared: 0.03595
## F-statistic: 63.34 on 1 and 1671 DF, p-value: 3.1755e-15
```

Although the coefficients are statistically significant (at the 1% significance level) for this model, they are extremely small, so is the R-squared value. The trend component does not improve the model's explanatory power by much and is therefore not considered for further modelling of `data2`.

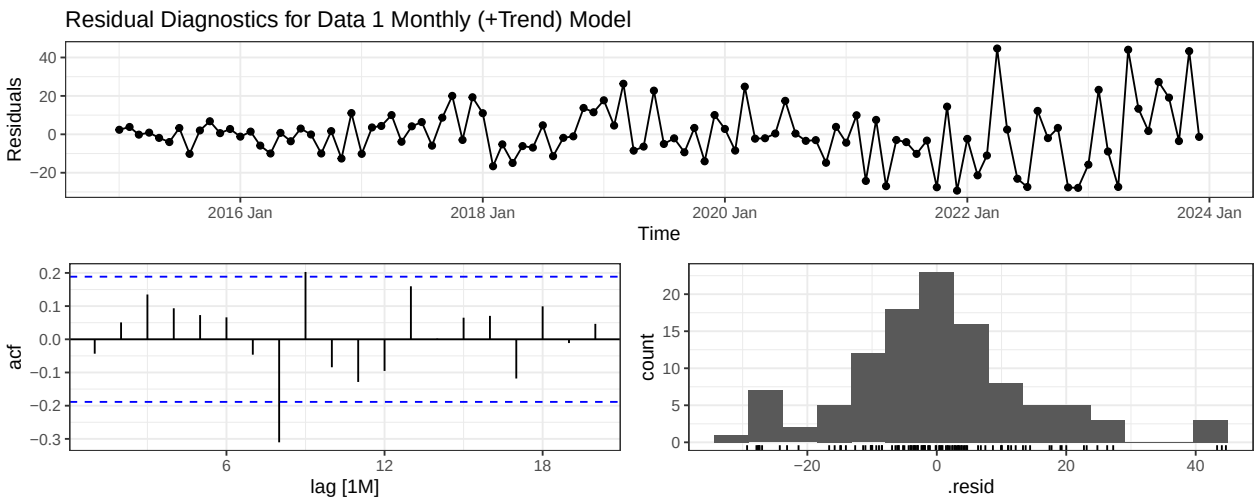
II. Assessing Seasonality

- *Check the need for seasonal fixed effects in the residuals of the model(s) produced in the previous point.*
- Multiple seasonal fixed effects might be needed. Check them in the order discussed in class.

Series 1

```
data1_monthly_dummies = data1_final %>% mutate(month = factor(month(t)))

data1_monthly_dummies_trend <- data1_monthly_dummies%>%
  model(TSLM(value ~ trend() + month))
```

```
gg_tsresiduals(data1_monthly_dummies_trend) +
labs(title = "Residual Diagnostics for Data 1 Monthly (+Trend) Model",
       x = "Time", y = "Residuals")
```



The residuals of the trend model incorporating monthly seasonality display residuals mostly centered around 0 in the time series of the residuals as well as the histogram. The ACF has two significant lags (8 and 9) and does not show any further pattern. This is comparable to the residual analysis for the trend model without incorporating seasonality.

```
data1_monthly_dummies_trend %>%
  report()
```

```
## Series: value
## Model: TSLM
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -29.310  -7.300  -1.135   5.116  44.669
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.33655    5.61176   0.060   0.9523
## trend()      0.95136    0.04756  20.002   <2e-16 ***
## month2       5.96143    7.21963   0.826   0.4110
## month3       2.84969    7.22010   0.395   0.6940
## month4       4.50117    7.22088   0.623   0.5345
## month5       4.87627    7.22198   0.675   0.5012
## month6       3.77011    7.22339   0.522   0.6029
## month7       0.59996    7.22511   0.083   0.9340
## month8       4.32333    7.22715   0.598   0.5511
## month9       8.76765    7.22949   1.213   0.2282
## month10     -2.80566    7.23215  -0.388   0.6989
```

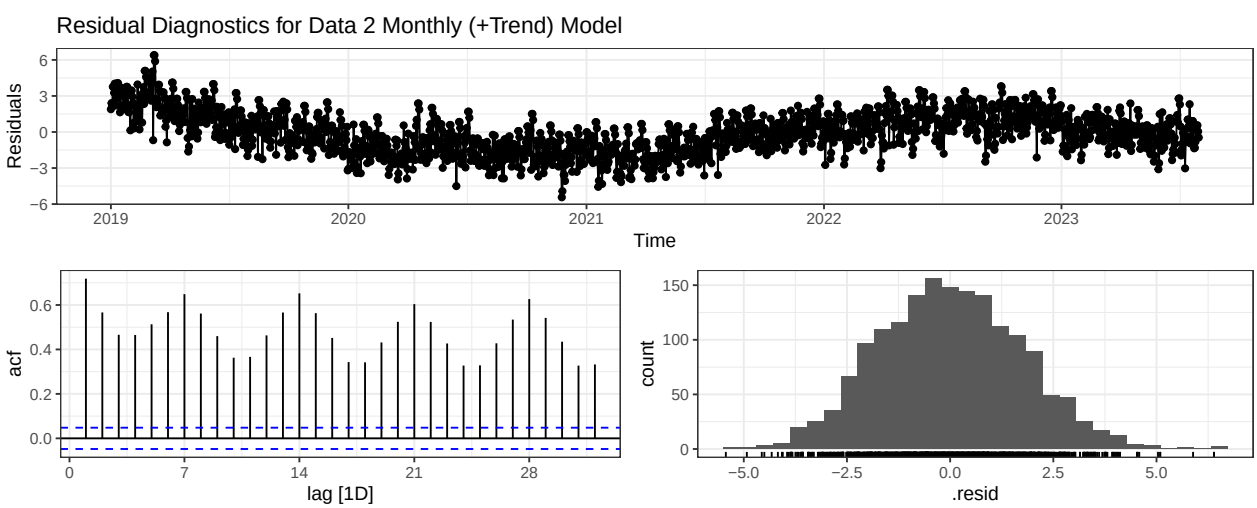
```
## month11     15.20702    7.23512   2.102   0.0382 *
## month12      3.36893    7.23841   0.465   0.6427
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.31 on 95 degrees of freedom
## Multiple R-squared: 0.8146,  Adjusted R-squared: 0.7912
## F-statistic: 34.78 on 12 and 95 DF, p-value: < 2.22e-16
```

The coefficients for month 1 (intercept after controlling trend) and each of the monthly dummies are not statistically significant. Only month 11, which has a value of 15.02 higher that the baseline while controlling for trend, is significant at the 10% level and could be considered to be incorporated. Although the R-squared is high, this is almost solely to the highly significant `trend()` component of the model. The lack of statistical significance for most dummies suggests that monthly seasonality might not be the best way to model periodic effects.

Series 2

```r
data2_dummies <- data2 %>%
  mutate(
    month = factor(month(t)),
    day_of_week = factor(wday(t))
  )

data2_dummies_trend_month <- data2_dummies %>%
  model(TSLM(u ~ trend() + month))

data2_dummies_trend_day <- data2_dummies %>%
  model(TSLM(u ~ trend() + day_of_week))

data2_dummies_trend_month_day <- data2_dummies %>%
  model(TSLM(u ~ trend() + month + day_of_week))
```

```r
gg_tsresiduals(data2_dummies_trend_month) +
  labs(title = "Residual Diagnostics for Data 2 Monthly (+Trend) Model",
       x = "Time", y = "Residuals")
```



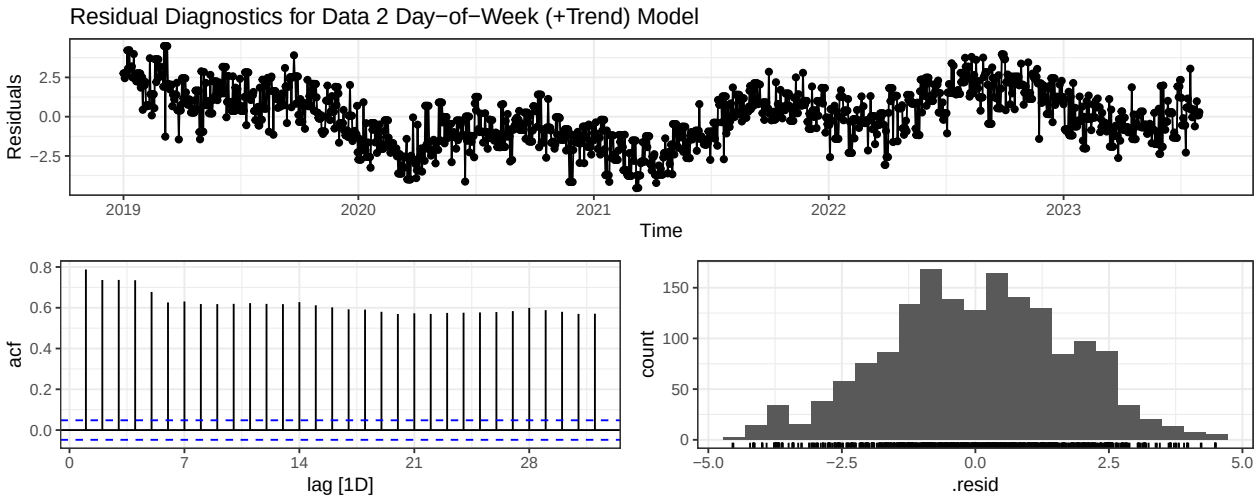
The residual analysis shows similar visualizations as for the trend model without monthly dummies. Just as before, this suggests that the model is still not capturing all seasonal components present in `data2`.

```r
data2_dummies_trend_month%>%
  report()
```

```
## Series: u
## Model: TSLM
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.4347 -1.1795 -0.0419  1.2237  6.3928
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.476e+01  1.538e-01 160.976  < 2e-16 ***
## trend()     -7.617e-04  8.819e-05  -8.637  < 2e-16 ***
## month2      -4.305e-01  2.014e-01  -2.137 0.032726 *
## month3      -8.879e-01  1.967e-01  -4.515 6.78e-06 ***
## month4      -6.672e-01  1.984e-01  -3.363 0.000788 ***
## month5      -9.097e-02  1.969e-01  -0.462 0.644093
## month6       1.902e-01  1.987e-01   0.957 0.338627
## month7       5.847e-01  1.972e-01   2.964 0.003078 **
## month8       8.200e-01  2.085e-01   3.932 8.76e-05 ***
## month9       1.194e+00  2.105e-01   5.674 1.64e-08 ***
## month10      9.184e-01  2.087e-01   4.401 1.15e-05 ***
## month11      4.384e-01  2.107e-01   2.080 0.037637 *
## month12      7.628e-02  2.089e-01   0.365 0.715114
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.731 on 1660 degrees of freedom
## Multiple R-squared: 0.1443,  Adjusted R-squared: 0.1382
## F-statistic: 23.34 on 12 and 1660 DF, p-value: < 2.22e-16
```

The coefficients of several monthly dummies are significant at the 1% level, while `month7` is significant at the 5% level and `month2` and `month11` at the 10% level. The R-squared increased by more than 10% points by adding monthly dummies, highlighting that monthly seasonality dummies has a substantial impact on how much variance in the data is explained.

```r
gg_tsresiduals(data2_dummies_trend_day) +
  labs(title = "Residual Diagnostics for Data 2 Day-of-Week (+Trend) Model",
       x = "Time", y = "Residuals")
```



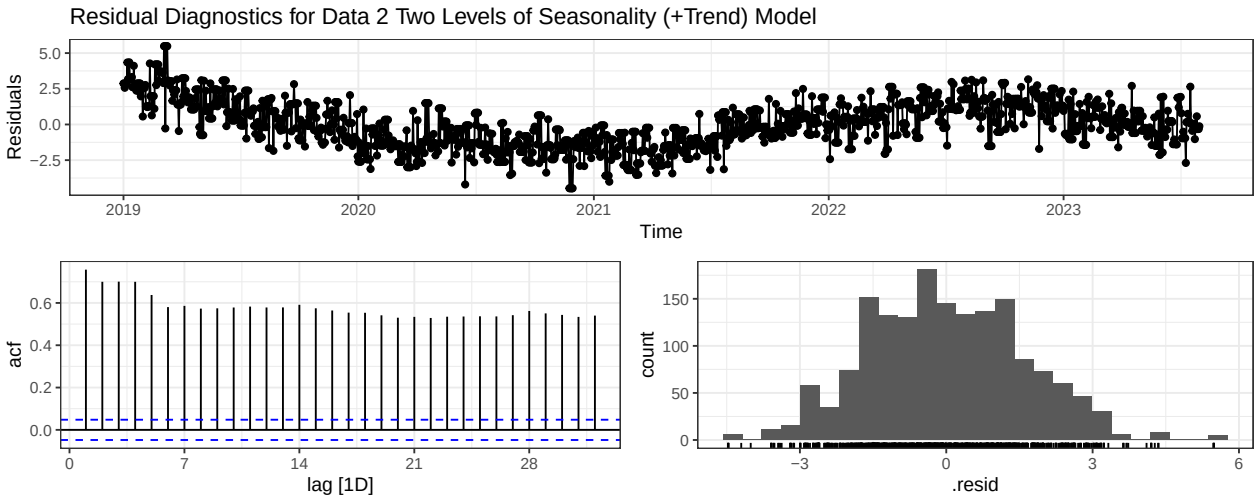
This residual analysis for the trend model that incorporates day of week dummies shows that the residuals are centered around 0 but exhibit a clear pattern, indicating that some structure remains unexplained (potentially the monthly seasonality from before or something else). The ACF plot reveals significant autocorrelations across multiple lags, suggesting the model does not fully account for temporal dependencies. The histogram of residuals is approximately bell-shaped, indicating near-normality but with potential room for improvement in the model's fit.

```
data2_dummies_trend_day%>%
  report()
```

```
## Series: u
## Model: TSLM
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -4.547468 -1.191299 -0.003222  1.245522  4.496973
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.527e+01  1.319e-01 191.584  < 2e-16 ***
## trend()      -7.406e-04  8.627e-05  -8.584  < 2e-16 ***
## day_of_week2 -6.956e-01  1.559e-01  -4.462 8.66e-06 ***
## day_of_week3 -1.364e+00  1.559e-01  -8.751  < 2e-16 ***
## day_of_week4 -8.541e-01  1.559e-01  -5.479 4.94e-08 ***
## day_of_week5 -8.044e-01  1.559e-01  -5.160 2.77e-07 ***
## day_of_week6  5.050e-01  1.559e-01   3.239  0.00122 **
## day_of_week7  5.099e-01  1.559e-01   3.270  0.00110 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.704 on 1665 degrees of freedom
## Multiple R-squared: 0.1678,  Adjusted R-squared: 0.1643
## F-statistic: 47.97 on 7 and 1665 DF, p-value: < 2.22e-16
```

The model shows significant coefficients for all week day dummies. The R-squared has also increased by approximately 12 percentage points compared to the model which only captures trend.

```
gg_tsresiduals(data2_dummies_trend_month_day)+
    labs(title = "Residual Diagnostics for Data 2 Two Levels of Seasonality (+Trend) Model",
       x = "Time", y = "Residuals")
```



The residual diagnostics for the combined levels of seasonality dummies model indicate improvements compared to individual models. However, the residuals still show structured patterns and significant autocorrelations remain in the ACF. This suggests further adjustments may be needed to fully capture dependencies based on time.

```
data2_dummies_trend_month_day%>%
  report()
```

```
## Series: u
## Model: TSLM
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.47004 -1.20928 -0.09817  1.11077  5.48380
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.515e+01  1.708e-01 147.259  < 2e-16 ***
## trend()      -7.646e-04  8.132e-05  -9.402  < 2e-16 ***
## month2       -4.343e-01  1.857e-01  -2.338 0.019495 *
## month3       -8.702e-01  1.813e-01  -4.799 1.74e-06 ***
## month4       -6.736e-01  1.829e-01  -3.682 0.000239 ***
## month5       -9.132e-02  1.815e-01  -0.503 0.615014
## month6        2.055e-01  1.832e-01   1.122 0.262212
## month7        5.699e-01  1.819e-01   3.133 0.001759 **
## month8        8.252e-01  1.923e-01   4.291 1.88e-05 ***
## month9        1.212e+00  1.941e-01   6.244 5.42e-10 ***
## month10       8.998e-01  1.924e-01   4.676 3.16e-06 ***
## month11       4.484e-01  1.943e-01   2.308 0.021123 *
## month12       9.036e-02  1.927e-01   0.469 0.639138
## day_of_week2 -6.972e-01  1.460e-01  -4.776 1.95e-06 ***
## day_of_week3 -1.358e+00  1.460e-01  -9.303  < 2e-16 ***
## day_of_week4 -8.500e-01  1.460e-01  -5.823 6.95e-09 ***
## day_of_week5 -8.038e-01  1.460e-01  -5.505 4.26e-08 ***
## day_of_week6  5.081e-01  1.460e-01   3.480 0.000514 ***
## day_of_week7  5.094e-01  1.460e-01   3.489 0.000497 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.596 on 1654 degrees of freedom
## Multiple R-squared: 0.2752,  Adjusted R-squared: 0.2673
## F-statistic: 34.88 on 18 and 1654 DF, p-value: < 2.22e-16
```

The model combining trend, month, and day dummies shows multiple statistically significant coefficients for both seasonal and trend components. The R-squared reflects better explanatory power compared to simpler models, but there is still room for improvement (as also indicated in the residual analysis).

III. Include Events

- *Including events - Christmas day, as well as for the day before and the day after.*

```r
data2 %>%
  mutate(
    month = factor(month(t)),
    day_of_week = factor(wday(t)),
    christmas = day(t) == 25 & month(t) == 12,
    day_before_christmas = day(t) == 24 & month(t) == 12,
    day_after_christmas = day(t) == 26 & month(t) == 12
  ) %>%
  relocate(t, u, christmas, day_before_christmas, day_after_christmas) -> data2_christmas

data2_model_christmas <- data2_christmas %>%
  model(TSLM(u ~ trend() + month + day_of_week + christmas + day_before_christmas + day_after_christmas

data2_model_christmas%>%
  report()
```

```
## Series: u
## Model: TSLM
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.46970 -1.20945 -0.09837  1.11061  5.48352
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               2.515e+01  1.709e-01 147.125  < 2e-16 ***
## trend()                  -7.646e-04  8.138e-05  -9.396  < 2e-16 ***
## month2                   -4.343e-01  1.859e-01  -2.336 0.019595 *
## month3                   -8.702e-01  1.815e-01  -4.795 1.77e-06 ***
## month4                   -6.735e-01  1.831e-01  -3.679 0.000242 ***
## month5                   -9.132e-02  1.817e-01  -0.503 0.615289
## month6                    2.055e-01  1.833e-01   1.121 0.262574
## month7                    5.699e-01  1.820e-01   3.131 0.001773 **
## month8                    8.252e-01  1.924e-01   4.288 1.91e-05 ***
## month9                    1.212e+00  1.943e-01   6.239 5.59e-10 ***
## month10                   8.999e-01  1.926e-01   4.673 3.21e-06 ***
## month11                   4.484e-01  1.945e-01   2.306 0.021226 *
## month12                   6.559e-02  1.985e-01   0.330 0.741088
## day_of_week2             -6.970e-01  1.461e-01  -4.769 2.01e-06 ***
## day_of_week3             -1.358e+00  1.462e-01  -9.288  < 2e-16 ***
## day_of_week4             -8.485e-01  1.461e-01  -5.806 7.66e-09 ***
```

```
## day_of_week5             -8.051e-01  1.462e-01  -5.507 4.23e-08 ***
## day_of_week6              5.080e-01  1.462e-01   3.475 0.000524 ***
## day_of_week7              5.077e-01  1.461e-01   3.474 0.000526 ***
## christmasTRUE             3.869e-02  8.134e-01   0.048 0.962069
## day_before_christmasTRUE  3.941e-01  8.134e-01   0.484 0.628103
## day_after_christmasTRUE   3.357e-01  8.134e-01   0.413 0.679902
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.597 on 1651 degrees of freedom
## Multiple R-squared: 0.2753,  Adjusted R-squared: 0.2661
## F-statistic: 29.87 on 21 and 1651 DF, p-value: < 2.22e-16
```

None of the coefficients of the special day dummies (Christmas, day before and after) incorporated in this model are statistically significant. Therefore, their inclusion does not improve the model.

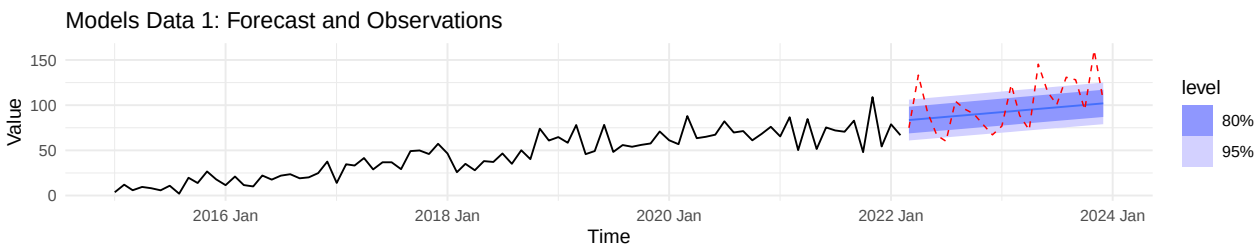
IV Predictions & accuracy

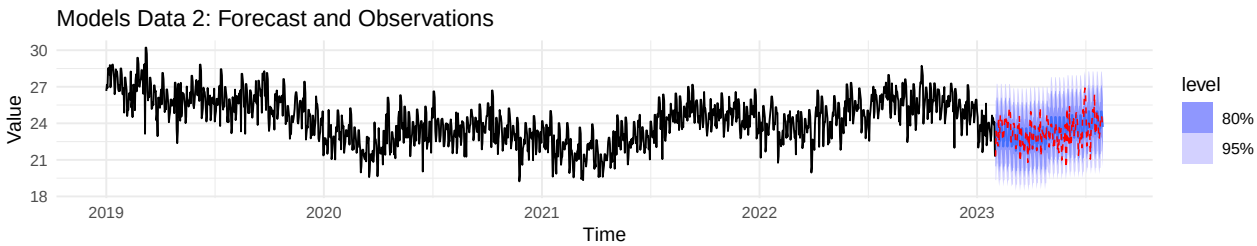
- Split the data into a training and a testing set, make predictions, visualize, and compute accuracy metrics.
- *Do this in the same way as in Assignment 1: follow those instructions.*
- *Include the benchmark models (only in this part, no need to include them during model building).*
- *If you have reduced any of the data and the split is a percentage, you can apply the percentage to the reduced data, rounding as needed.*

```r
data2_dummies <- data2 %>%
  mutate(
    month = factor(month(t)),
    day_of_week = factor(wday(t))
  )

split_index <- round(0.8 * nrow(data1_final))
train_data1 <- data1_final %>% filter(row_number() <= split_index)
test_data1 <- data1_final %>% filter(row_number() > split_index)

cutoff_date <- as.Date("2023-02-01")
test_data2 <- data2_dummies %>% filter(t >= cutoff_date)
train_data2 <- data2_dummies %>% filter(t < cutoff_date)

# All-in-One Forecast Models
# Data1
horizon_data1 <- nrow(test_data1)
forecasts_all_data_1 <- train_data1 %>%
  model(
    Naive = NAIVE(value),
    Mean = MEAN(value),
    Snaive = SNAIVE(value),
    Drift = RW(value ~ drift()),
    Trend = TSLM(value ~ trend()) # New Model
  ) %>%
  forecast(h = horizon_data1)

# Data2
horizon_data2 <- nrow(test_data2)
```

```r
forecasts_all_data2 <- train_data2 %>%
  model(
    Naive = NAIVE(u),
    Mean = MEAN(u),
    Snaive = SNAIVE(u),
    Drift = RW(u ~ drift()),
    TSLM(u ~ trend() + month + day_of_week)
  ) %>%
  forecast(new_data = test_data2)
```

```r
Model_data1 <-train_data1 %>%
  model(TSLM(value ~ trend()))%>%
  forecast(h = horizon_data1)

Model_data2 <-train_data2%>%
  model(TSLM(u ~ trend() + month + day_of_week))%>%
  forecast(new_data = test_data2)

grid.arrange(
  autoplot(Model_data1, train_data1) +
    autolayer(test_data1, colour = "red", size = 0.4, linetype = "dashed") +
    labs(
      title = "Models Data 1: Forecast and Observations",
      x = "Time",
      y = "Value"
    ) +
    theme_minimal(),

  autoplot(Model_data2, train_data2) +
    autolayer(test_data2, colour = "red", size = 0.4, linetype = "dashed") +
    labs(
      title = "Models Data 2: Forecast and Observations",
      x = "Time",
      y = "Value"
    ) +
    theme_minimal(),
  ncol = 1
)
```






```
accuracy_metrics1 <- forecasts_all_data_1 %>%
  accuracy(test_data1)

accuracy_metrics2 <- forecasts_all_data2 %>%
  accuracy(test_data2)
```

```
accuracy_metrics1
```

```
## # A tibble: 5 x 10
##   .model .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE     ACF1
##   <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>    <dbl>
## 1 Drift  Test  24.9   35.2  27.1 20.0   23.5   NaN   NaN -0.0215
## 2 Mean   Test  54.9   61.1  54.9 51.5   51.5   NaN   NaN  0.109
## 3 Naive  Test  33.4   42.9  34.0 28.5   29.5   NaN   NaN  0.109
## 4 Snaive Test  30.0   41.6  36.0 25.5   33.8   NaN   NaN  0.00699
## 5 Trend  Test   7.40  25.7  20.2  1.41  19.7   NaN   NaN -0.0430
```

The Trend model achieves the best accuracy, with the lowest root mean square error (RMSE) of 25.72 and mean absolute error (MAE) of 20.18, demonstrating its superior ability to capture the underlying trends in `data1` compared to baseline models.

```
accuracy_metrics2
```

```
## # A tibble: 5 x 10
##   .model                .type       ME  RMSE   MAE    MPE  MAPE  MASE RMSSE  ACF1
##   <chr>                 <chr>    <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Drift                 Test   2.40     2.69 2.41  10.1   10.1    NaN   NaN 0.475
## 2 Mean                  Test  -1.02     1.55 1.27  -4.65   5.61   NaN   NaN 0.438
## 3 Naive                 Test   2.07     2.38 2.10   8.66   8.80   NaN   NaN 0.438
## 4 Snaive                Test  -0.355    1.22 0.984 -1.66   4.27   NaN   NaN 0.265
## 5 TSLM(u ~ trend() + ~ Test   0.00487  1.02 0.802 -0.132  3.44   NaN   NaN 0.372
```

The Trend model incorporating trend, monthly, and weekday effects achieves the highest accuracy for `data2`compared to other simple models, with the lowest root mean square error (RMSE) of 1.02 and mean absolute error (MAE) of 0.80.

```
tibble(
  .model = forecasts_all_data_1$.model,
  fc_error = forecasts_all_data_1$.mean - test_data1$value ) %>%
  group_by(.model) %>%
  summarise(MAD = median(abs(fc_error), na.rm = TRUE))
```

```
## # A tibble: 5 x 2
##   .model   MAD
##   <chr>  <dbl>
## 1 Drift   21.7
## 2 Mean    49.9
## 3 Naive   28.5
## 4 Snaive  38.6
## 5 Trend   17.1
```

The Trend model demonstrates the best performance for `data1`, with a median absolute deviation (MAD) of 17.10.

```
tibble(
  .model = forecasts_all_data2$.model,
  fc_error = forecasts_all_data2$.mean- test_data2$u)%>%
  group_by(.model)%>%
    summarise(MAD = median(abs(fc_error), na.rm = TRUE))
```

```
## # A tibble: 5 x 2
##   .model                                    MAD
##   <chr>                                   <dbl>
## 1 Drift                                   2.40
## 2 Mean                                    1.12
## 3 Naive                                   2.00
## 4 Snaive                                  0.783
## 5 TSLM(u ~ trend() + month + day_of_week) 0.605
```

For `data2`, the combined model with trend, monthly, and day-of-week effects outperforms all others, achieving the lowest median absolute deviation (MAD) of 0.60.